

HDOC Day- 14

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

Check whether all its digits are unique.

E.g. 1: Number: 12345, Output: Unique

E.g. 2: Number: 12234, Output: Not Unique

Find the frequency of each digit from '0' to '9'.

E.g. Number: 112333, freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3, freq_4 = 0, freq_5 = 0, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0

AIM

To write a menu-driven C program using switch-case to perform different operations on the digits of a positive integer, such as checking whether all digits are unique and finding the frequency of each digit from 0 to 9.

Algorithm

1. Start.
2. Enter a positive integer n.
3. Display the menu:
 - 1. Check whether all digits are unique.
 - 2. Find frequency of each digit.
4. Enter the choice.
5. Use switch statement for the choice.
6. **If choice is 1:**
 - Take an array freq[10] and initialize it with 0.
 - Extract each digit using $n \% 10$.
 - Check whether the digit has already occurred.
 - If yes, print "**Not Unique**".
 - Otherwise, continue checking all digits.
 - If no digit is repeated, print "**Unique**".

7. If choice is 2:

- Take an array freq[10] and initialize it with 0.
- Extract each digit one by one.
- Increase the frequency of that digit.
- Print the frequency of digits from 0 to 9.

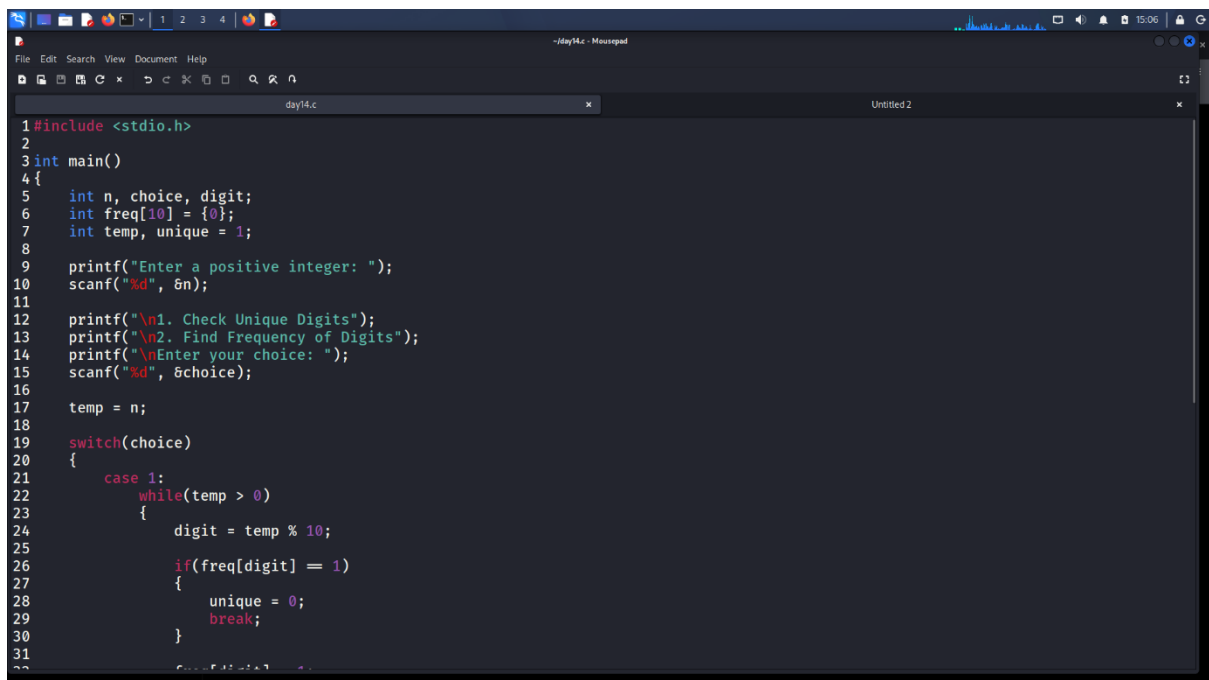
8. If the choice is not 1 or 2, print "**Invalid Choice**".

9. Stop.

Test Case Input Number Choice Expected Output

1	12345	1	Unique
2	12234	1	Not Unique
3	112233	2	1 = 2, 2 = 2, 3 = 2
4	9876	2	9 = 1, 8 = 1, 7 = 1, 6 = 1
5	123	3	Invalid Choice

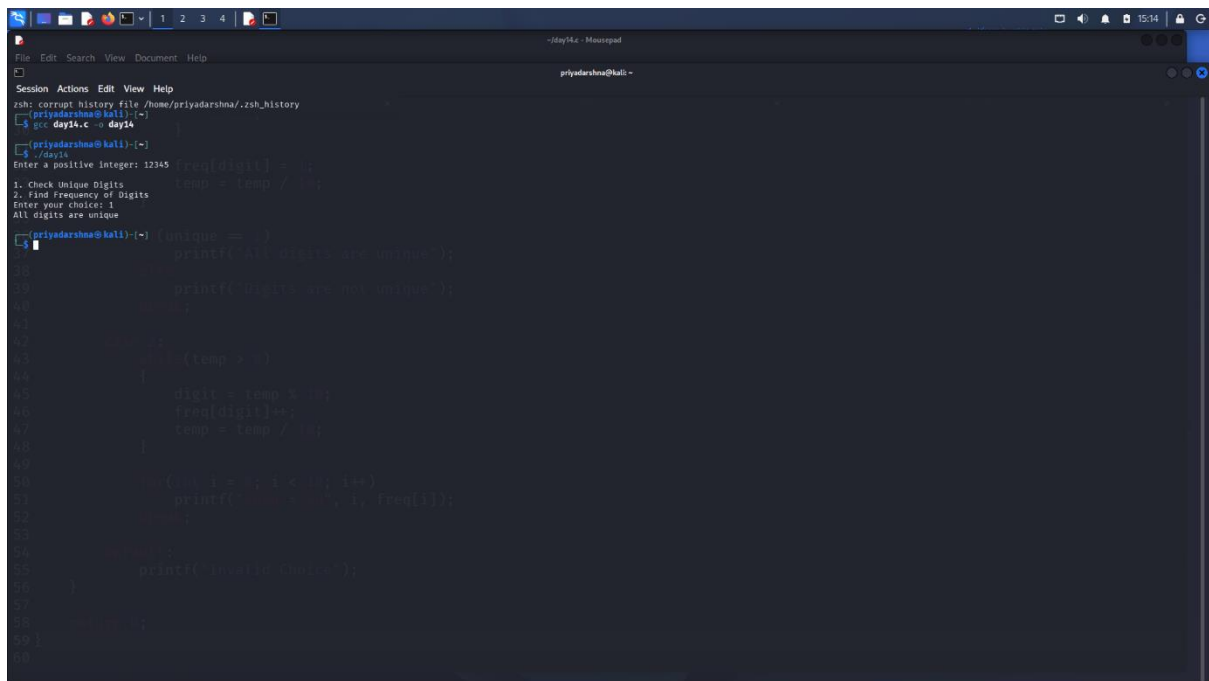
>Write a C program code

A screenshot of a code editor window titled "day14.c" showing a C program. The program includes <stdio.h> and defines a main function. It prompts the user to enter a positive integer and a choice. Based on the choice, it either checks for unique digits (choice 1) or finds the frequency of digits (choice 2). For choice 1, it uses a while loop to extract digits and checks if they are unique. For choice 2, it uses an array to store the frequency of each digit from 0 to 9. The code is as follows:

```
1#include <stdio.h>
2
3int main()
4{
5    int n, choice, digit;
6    int freq[10] = {0};
7    int temp, unique = 1;
8
9    printf("Enter a positive integer: ");
10   scanf("%d", &n);
11
12   printf("\n1. Check Unique Digits");
13   printf("\n2. Find Frequency of Digits");
14   printf("\nEnter your choice: ");
15   scanf("%d", &choice);
16
17   temp = n;
18
19   switch(choice)
20   {
21       case 1:
22           while(temp > 0)
23           {
24               digit = temp % 10;
25
26               if(freq[digit] == 1)
27               {
28                   unique = 0;
29                   break;
30               }
31           }
32       case 2:
33           while(temp > 0)
34           {
35               digit = temp % 10;
36               freq[digit]++;
37               temp /= 10;
38           }
39           for(int i = 0; i < 10; i++)
40           {
41               printf("%d : %d\n", i, freq[i]);
42           }
43       default:
44           printf("Invalid Choice");
45   }
46 }
```

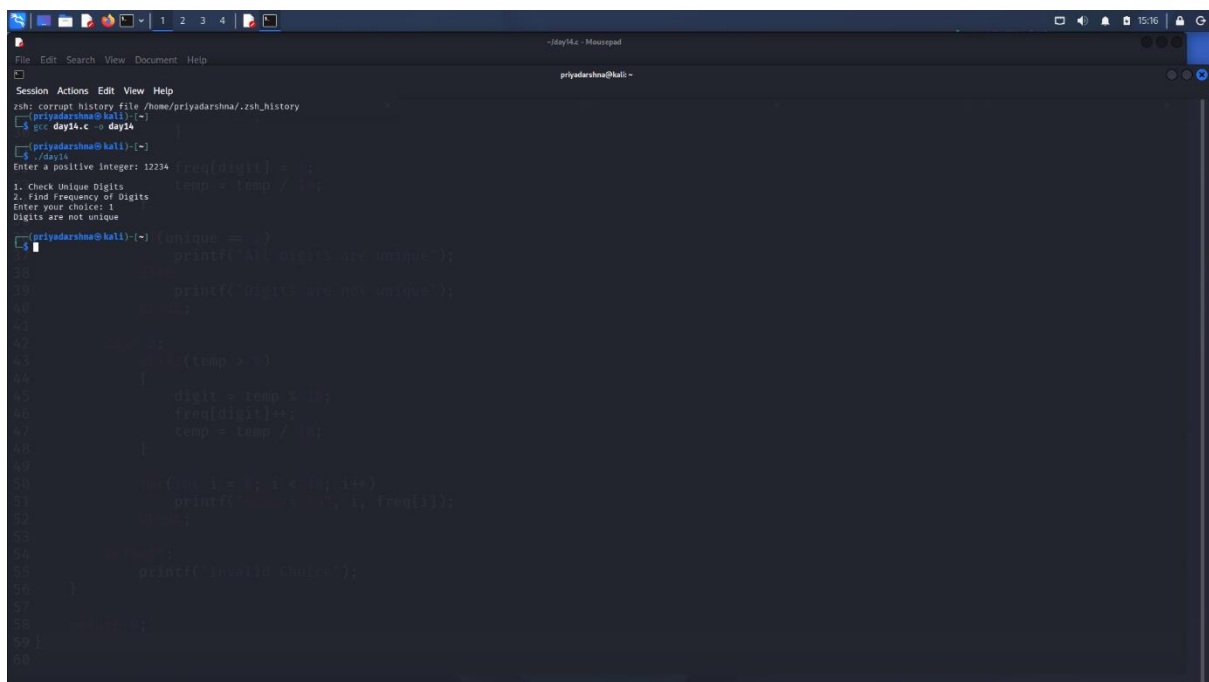
```
30     }
31
32     freq[digit] = 1;
33     temp = temp / 10;
34 }
35
36 if(unique == 1)
37     printf("All digits are unique");
38 else
39     printf("Digits are not unique");
40 break;
41
42 case 2:
43     while(temp > 0)
44     {
45         digit = temp % 10;
46         freq[digit]++;
47         temp = temp / 10;
48     }
49
50     for(int i = 0; i < 10; i++)
51         printf("\n%d = %d", i, freq[i]);
52     break;
53
54 default:
55     printf("Invalid Choice");
56 }
57
58 return 0;
59 }
60
```

>Test Case 1-All the digits are unique.



```
File Edit Search View Document Help
~day14 - Mousepad
priyadarshna@kali -
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
~(priyadarshna@kali)~
$ gcc day14.c -o day14
~(priyadarshna@kali)~
$ ./day14
Enter a positive integer: 12345
1. Check Unique Digits
2. Find Frequency of Digits
Enter your choice: 1
All digits are unique
~(priyadarshna@kali)~
1 (unique == 1)
printf("All digits are unique");
2
printf("Digits are not unique");
3
return 0;
4
}
5
int main()
6 {
7     int temp = 10;
8     int digit;
9     int freq[10] = {0};
10    while(temp > 0)
11    {
12        digit = temp % 10;
13        freq[digit]++;
14        temp = temp / 10;
15    }
16    for(int i = 0; i < 10; i++)
17    {
18        printf("%d : %d\n", i, freq[i]);
19    }
20    return 0;
21 }
```

>Test case- 2 Digits are not unique.



```
File Edit Search View Document Help
~day14 - Mousepad
priyadarshna@kali -
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
~(priyadarshna@kali)~
$ gcc day14.c -o day14
~(priyadarshna@kali)~
$ ./day14
Enter a positive integer: 12234
1. Check Unique Digits
2. Find Frequency of Digits
Enter your choice: 1
Digits are not unique
~(priyadarshna@kali)~
1 (unique == 0)
printf("All digits are unique");
2
printf("Digits are not unique");
3
return 0;
4
}
5
int main()
6 {
7     int temp = 10;
8     int digit;
9     int freq[10] = {0};
10    while(temp > 0)
11    {
12        digit = temp % 10;
13        freq[digit]++;
14        temp = temp / 10;
15    }
16    for(int i = 0; i < 10; i++)
17    {
18        printf("%d : %d\n", i, freq[i]);
19    }
20    return 0;
21 }
```

>Test case -3 Find frequency of digits.

```
Terminal Emulator
Use the command line

Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day14.c -o day14
(priyadarshna@kali)~$ ./day14
Enter a positive integer: 112233
1. Check Unique Digits
2. Find Frequency of Digits
Enter your choice: 2
0 = 0
1 = 2
2 = 2
3 = 2
4 = 0
5 = 0
6 = 0
7 = 0
8 = 0
9 = 0
(priyadarshna@kali)~$
```

The screenshot shows a terminal window with a dark background. The user has compiled a C program named 'day14.c' into 'day14'. They then run the program and enter the integer '112233'. The program prompts the user to choose between two options: '1. Check Unique Digits' and '2. Find Frequency of Digits'. The user selects option 2. The program then displays the frequency of each digit from 0 to 9. The output shows that digits 1, 2, and 3 each appear 2 times, while all other digits (0, 4, 5, 6, 7, 8, 9) appear 0 times.

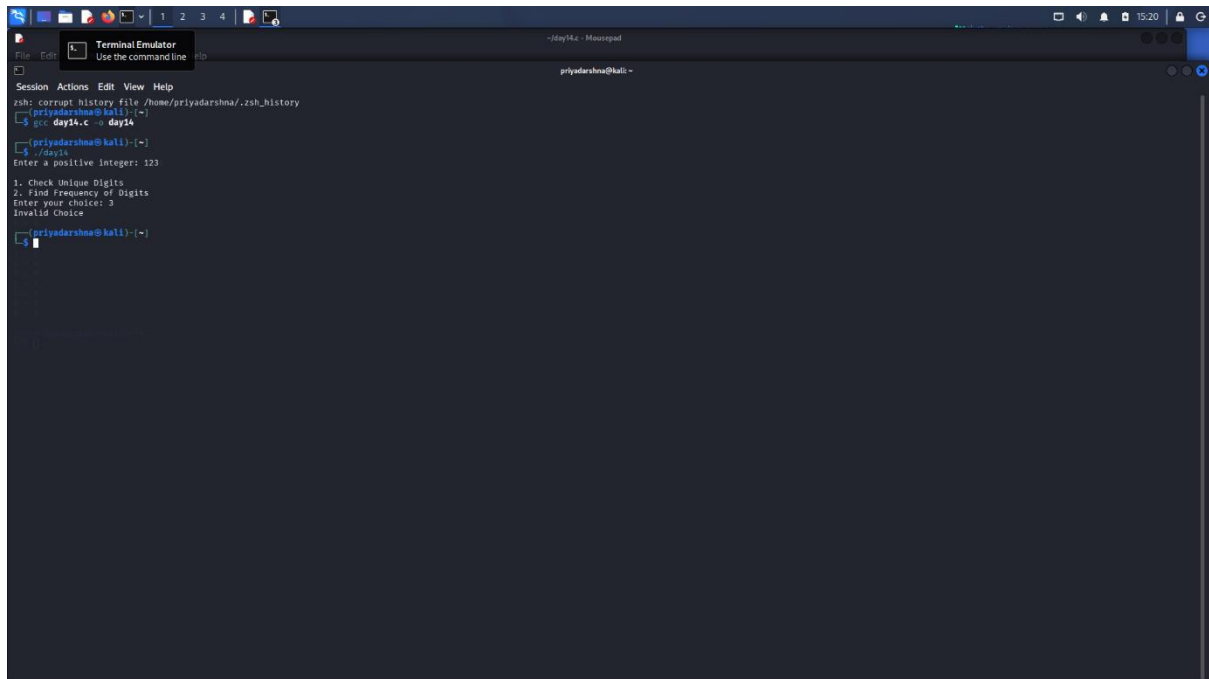
>Test Case -4 Digit Frequency test.

```
Terminal Emulator
Use the command line

Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day14.c -o day14
(priyadarshna@kali)~$ ./day14
Enter a positive integer: 9876
1. Check Unique Digits
2. Find Frequency of Digits
Enter your choice: 2
0 = 0
1 = 0
2 = 0
3 = 0
4 = 0
5 = 0
6 = 1
7 = 1
8 = 1
9 = 1
(priyadarshna@kali)~$
```

The screenshot shows the same terminal window as before. The user enters the integer '9876'. They again select option 2 to find the frequency of digits. The program outputs the frequency for each digit from 0 to 9. The output shows that digits 6, 7, 8, and 9 each appear 1 time, while all other digits (0, 1, 2, 3, 4, 5) appear 0 times.

>Test case -5 Invalid choice.



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day14.c -o day14
(priyadarshna@kali)~$ ./day14
Enter a positive integer: 123
1. Check Unique Digits
2. Find Frequency of Digits
Enter your choice: 3
Invalid Choice
(priyadarshna@kali)~$
```